

SMART INDIA HACKATHON 2025



Vanni

- Problem Statement ID - **SIH25246**
- Problem Statement Title - **Simplified captions in multiple Indian languages**
- Theme - **Clean & Green Technology**
- PS Category - **Software**
- Team ID - **SIH-S-180**
- Team Name - **Bits & Bytes**



“Develop a real-time closed captioning solution with simplified captions in multiple Indian languages for accessibility and inclusivity of Deaf and hard-of-hearing individuals.”

Proposed Solution & How to address the problem

- **Offline Functionality:** Unlike most apps, **Live Captions** works fully offline, ensuring accessibility without internet.
- **Low Literacy Support:** The **Vanni** makes complex text understandable for users with low literacy.
- **Noise Filtering:** System sound detection minimises background noise and notifications, improving caption accuracy.
- **Multilingual & Fast:** Supports **15 Indian languages** with **low latency** and **high accuracy (>90%)**, outperforming many competitors.

Innovate & Uniqueness



Multi
language
support



Offline
Functionality



Text
Simplification



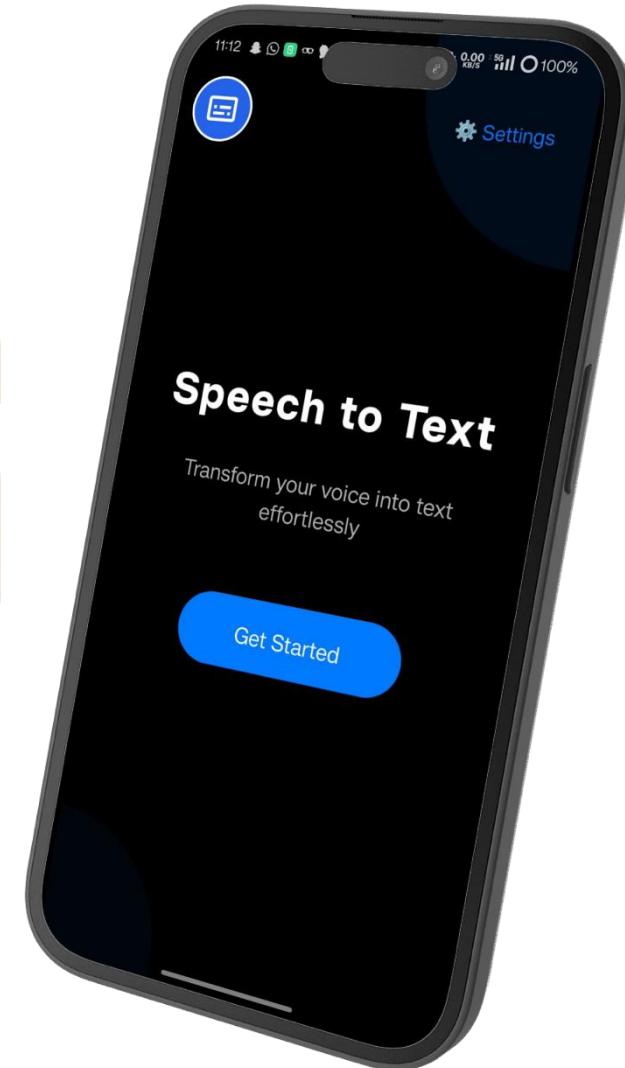
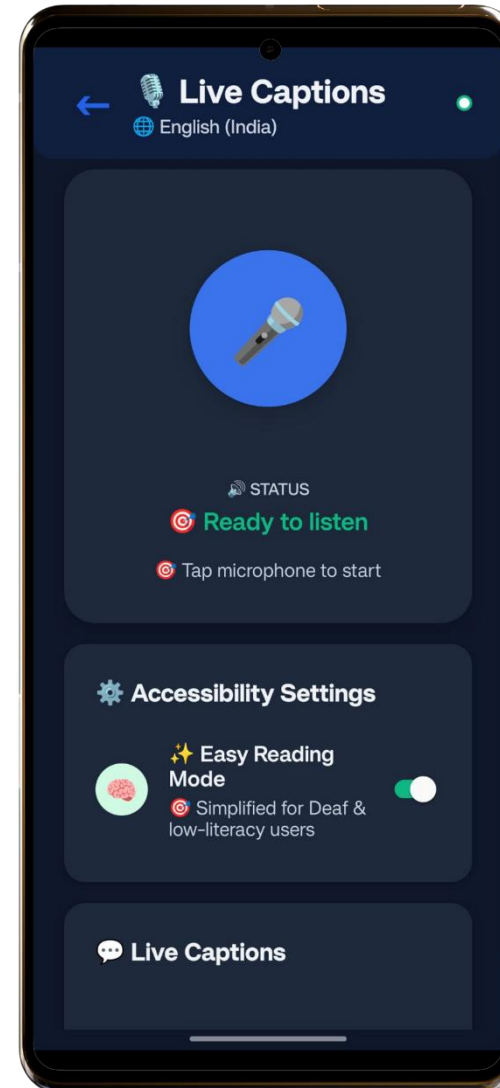
Voice
Commands



System Sound
Filter



Floating
Widget



Technology Used

User Interface:

- React Native
- TypeScript 5.0
- React Hooks

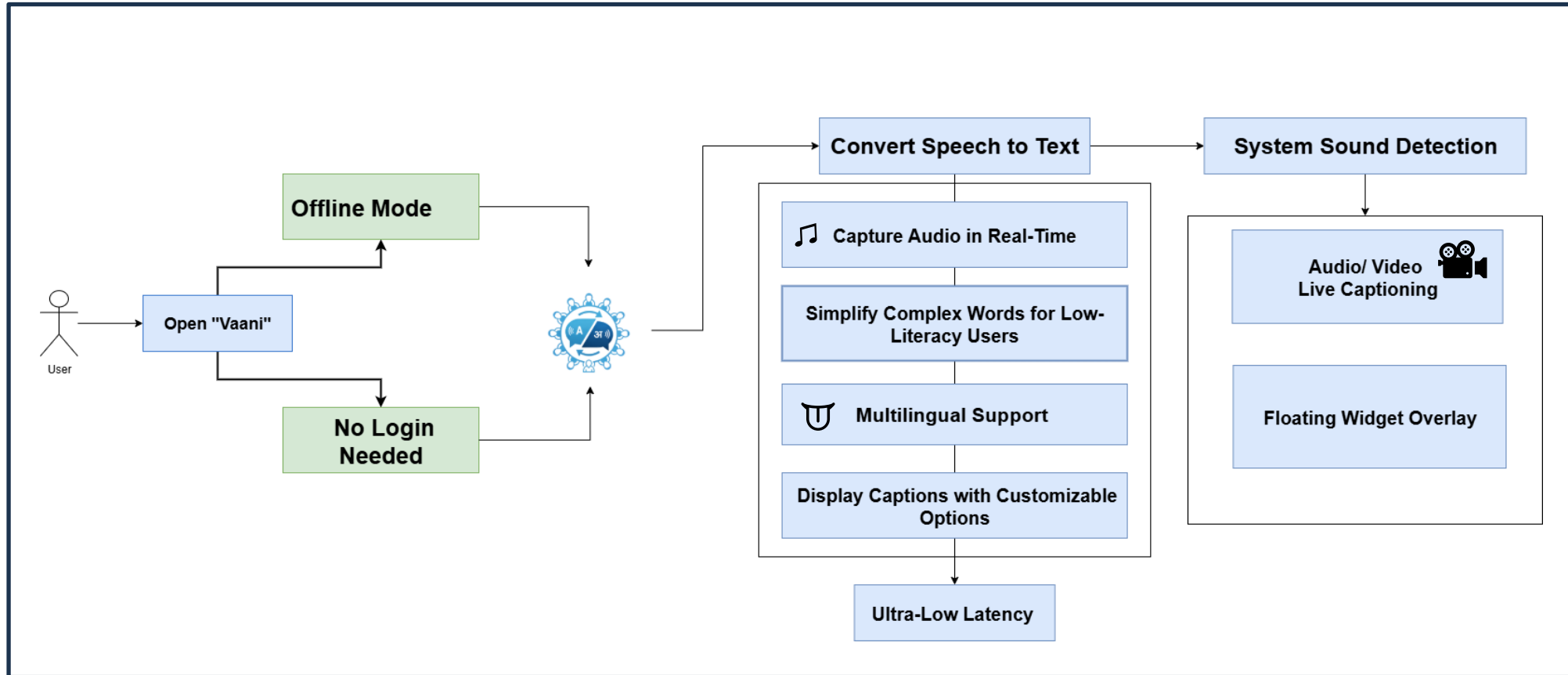


Speech Recognition:

- React-native-voice
 - AudioPlaybackCapture
- Handles real-time speech recognition, converting audio into text locally. 

- Android Studio (Android 10+)
- Used as the IDE for building, debugging, and testing Android apps. 

Work Flow Diagram



Feasibility Of the Idea

Low Maintenance

Reduced upkeep and cost-effective operation.

User-Friendly

Easy to use and implement.

Adaptable Architecture

Flexible design to suit changing requirements.



Sound detection

Audio files conversion from any app to Captions

Offline Fallback

Functionality available even without internet access.

Highly Scalable

Easily handle increased workloads and data.

Potential Challenges and Strategies for Overcoming



Floating Icon Overlay

A floating icon overlay across all applications.



Integrating voice commands

Complexity when functionality into React Native CodeBase.



Mobile App Compatibility

Mobile app compatibility across different Android and iOS device versions.



Verbal Communication Difficulties

Encounter difficulties in understanding verbal communication.



System Sounds Separation

System sounds and notifications separation.



React Native

Built on React Native, the app works offline without internet.



Single Codebase

React Native creates a single codebase for Android, iOS, and web.



Offline Mode

Offline mode ensures the app works even without connectivity.



Real-time Synchronization

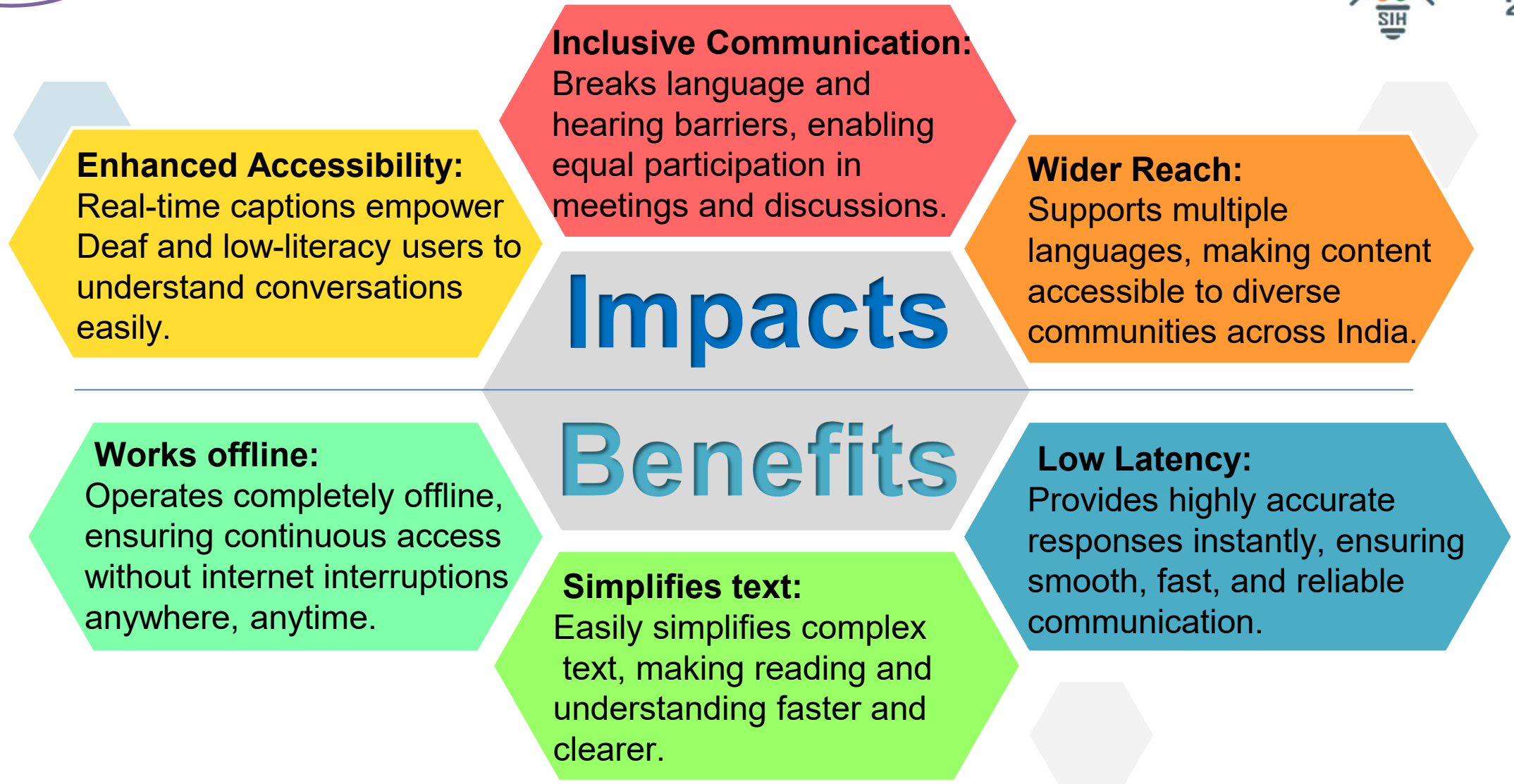
Real-time synchronization ensures low latency between speech and caption display.



User Feedback

User feedback mechanism to improve accuracy and simplify complex vocabulary continuously.

IMPACT AND BENEFITS



Evaluating Key Features

References



Feature/Aspect	Our App (Vanni)	Google Live Transcribe	Otter.ai
Offline Working	✓	✗	✗
Multilanguage Support	✓	✓	✗
Simplify Complex Words (Low literacy levels)	✓	✗	✗
Low Latency	✓	✓	✓
System Sound Detection	✓	✗	✗
High Accuracy	✓	✓	✓